

Homomorphisms

Road map

A homomorphism is a map that preserves multiplication and hence preserves the algebraic structure of a group. We develop its basic properties, images, preimages, kernel, injectivity criterion, and composition. We then distinguish endomorphisms, isomorphisms, and automorphisms, and prove again—now from the homomorphism viewpoint—that conjugation is an automorphism. Detailed examples include multiplication and reduction modulo m on $\mathbb{Z}$, left translations and the left regular representation, the sign map on S_n , the alternating group A_n , and the complete classification of cyclic groups.

6.1 Definition and first properties

Homomorphisms play the same structural role for groups that linear maps play for vector spaces: they are the maps compatible with the operations.

Definition 6.1: Group homomorphism

Let G and G' be groups. A map

$$\varphi : G \longrightarrow G'$$

is a *group homomorphism* if

$$\varphi(ab) = \varphi(a)\varphi(b) \quad \text{for all } a, b \in G.$$

The two products in this equation belong to different groups: ab is computed in G , while $\varphi(a)\varphi(b)$ is computed in G' . If the operations are denoted explicitly by $\circ$ on G and $\circ'$ on G' , the same condition is

$$\varphi(a \circ b) = \varphi(a) \circ' \varphi(b).$$

As in the handwritten notes, we usually use a deliberately strict notation and let context distinguish the two products.

Theorem 6.1: Basic properties of homomorphisms

Let $\varphi : G \rightarrow G'$ be a homomorphism, with identity elements $e \in G$ and $e' \in G'$. Then:

1. $\varphi(e) = e'$;

2. $\varphi(a^{-1}) = \varphi(a)^{-1}$ for every $a \in G$;
3. $\varphi(a^n) = \varphi(a)^n$ for every $a \in G$ and $n \in \mathbb{Z}$.

证明. Because $e = e^2$,

$$\varphi(e) = \varphi(e^2) = \varphi(e)\varphi(e).$$

Multiplying on the left by $\varphi(e)^{-1}$ gives $\varphi(e) = e'$.

Next,

$$e' = \varphi(e) = \varphi(aa^{-1}) = \varphi(a)\varphi(a^{-1}).$$

Thus $\varphi(a^{-1})$ is the inverse of $\varphi(a)$, so

$$\varphi(a^{-1}) = \varphi(a)^{-1}.$$

For $n > 0$, repeated use of the homomorphism property gives

$$\varphi(a^n) = \varphi(\underbrace{a \cdots a}_{n \text{ factors}}) = \underbrace{\varphi(a) \cdots \varphi(a)}_{n \text{ factors}} = \varphi(a)^n.$$

The case $n = 0$ is the identity calculation, and the result for $n < 0$ follows from the inverse calculation. ■

A consequence for element orders

If a has finite order n , then

$$\varphi(a)^n = \varphi(a^n) = \varphi(e) = e'.$$

Consequently, the order of $\varphi(a)$ divides the order of a . The order may decrease when φ is not injective.

6.2 Images and preimages of subgroups

Definition 6.2: Image and preimage

For a subset $H \subseteq G$, its *image* under φ is

$$\varphi(H) := \{\varphi(h) : h \in H\} \subseteq G'.$$

For a subset $H' \subseteq G'$, its *preimage* is

$$\varphi^{-1}(H') := \{a \in G : \varphi(a) \in H'\} \subseteq G.$$

The notation $\varphi^{-1}(H')$ denotes a preimage of a set and does not require φ to possess an inverse function.

Proposition 6.1: Subgroups pass through homomorphisms

Let $\varphi : G \rightarrow G'$ be a homomorphism.

1. If $H \leq G$, then $\varphi(H) \leq G'$.

2. If $H' \leq G'$, then $\varphi^{-1}(H') \leq G$.

证明. Suppose $H \leq G$. Since $e \in H$,

$$e' = \varphi(e) \in \varphi(H),$$

so the image is nonempty. Let $a', b' \in \varphi(H)$. There are $a, b \in H$ with $a' = \varphi(a)$ and $b' = \varphi(b)$. Then

$$\begin{aligned} a'(b')^{-1} &= \varphi(a)\varphi(b)^{-1} \\ &= \varphi(a)\varphi(b^{-1}) \\ &= \varphi(ab^{-1}). \end{aligned}$$

Because $H \leq G$, the element ab^{-1} belongs to H . Hence $a'(b')^{-1} \in \varphi(H)$, and the subgroup test proves $\varphi(H) \leq G'$.

Now suppose $H' \leq G'$. Since $\varphi(e) = e' \in H'$, the preimage is nonempty. Let $a, b \in \varphi^{-1}(H')$. Then $\varphi(a), \varphi(b) \in H'$, so

$$\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} \in H'.$$

Thus $ab^{-1} \in \varphi^{-1}(H')$, and the subgroup test gives $\varphi^{-1}(H') \leq G$. ■

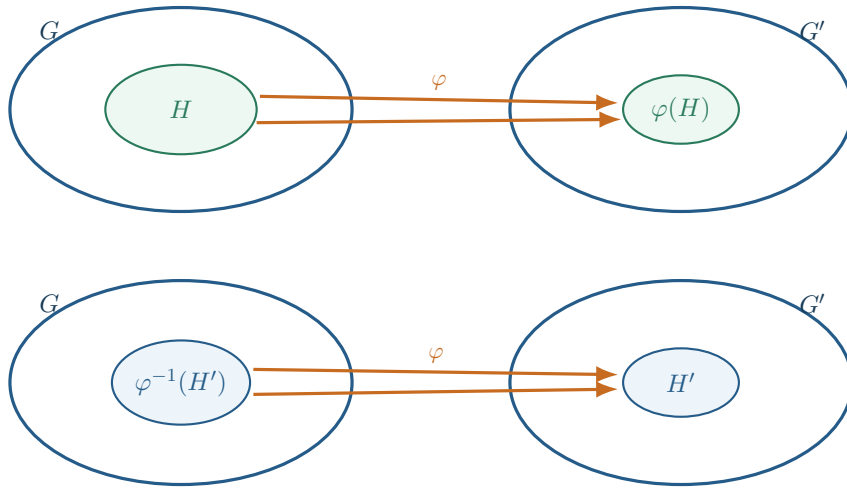


图 6.1: Top: the image $\varphi(H)$ of a subgroup $H \leq G$. Bottom: the full preimage $\varphi^{-1}(H')$ of a subgroup $H' \leq G'$. Both are subgroups.

6.3 Image, kernel, and injectivity

Definition 6.3: Image and kernel of a homomorphism

The *image* and *kernel* of $\varphi : G \rightarrow G'$ are

$$\text{im } \varphi := \{\varphi(a) : a \in G\} = \varphi(G)$$

and

$$\text{ker } \varphi := \{a \in G : \varphi(a) = e'\} = \varphi^{-1}(\{e'\}).$$

The previous proposition immediately gives

$$\text{im } \varphi \leq G', \quad \ker \varphi \leq G.$$

Indeed, these are the special cases $H = G$ and $H' = \{e'\}$.

Theorem 6.2: Kernel criterion for injectivity

A homomorphism $\varphi : G \rightarrow G'$ is injective if and only if

$$\ker \varphi = \{e\}.$$

证明. Suppose first that φ is injective. If $a \in \ker \varphi$, then

$$\varphi(a) = e' = \varphi(e).$$

Injectivity gives $a = e$. Thus $\ker \varphi = \{e\}$.

Conversely, suppose $\ker \varphi = \{e\}$, and assume $\varphi(a) = \varphi(b)$. Then

$$\begin{aligned} \varphi(a^{-1}b) &= \varphi(a)^{-1}\varphi(b) \\ &= \varphi(a)^{-1}\varphi(a) = e'. \end{aligned}$$

Therefore $a^{-1}b \in \ker \varphi = \{e\}$, so $a^{-1}b = e$, and hence $a = b$. Thus φ is injective. ■

Proposition 6.2: Composition

If

$$\varphi_1 : G \rightarrow G' \quad \text{and} \quad \varphi_2 : G' \rightarrow G''$$

are homomorphisms, then

$$\varphi_2 \circ \varphi_1 : G \rightarrow G''$$

is a homomorphism.

证明. For $a, b \in G$,

$$\begin{aligned} (\varphi_2 \circ \varphi_1)(ab) &= \varphi_2(\varphi_1(ab)) \\ &= \varphi_2(\varphi_1(a)\varphi_1(b)) \\ &= \varphi_2(\varphi_1(a))\varphi_2(\varphi_1(b)) \\ &= (\varphi_2 \circ \varphi_1)(a)(\varphi_2 \circ \varphi_1)(b). \end{aligned}$$

■

6.4 Endomorphisms, isomorphisms, and automorphisms

Definition 6.4: Three kinds of homomorphism

Let $\varphi : G \rightarrow G'$ be a homomorphism.

- It is an *endomorphism* if $G' = G$.
- It is an *isomorphism* if it is bijective.

- It is an *automorphism* if it is a bijective endomorphism, that is, if $G' = G$ and φ is bijective.

If an isomorphism $G \rightarrow G'$ exists, the groups are called *isomorphic*, and we write

$$G \cong G'.$$

Isomorphic groups have the same group-theoretic structure with different names for the elements. For finite groups, an isomorphism is a bijection, so

$$|G| = |G'|.$$

After corresponding elements are placed in the same order, their Cayley tables are identical.

Proposition 6.3

The inverse of an isomorphism is an isomorphism.

证明. Let $\varphi : G \rightarrow G'$ be an isomorphism. Its inverse function exists because φ is bijective. Take $x', y' \in G'$, and write $x' = \varphi(x)$, $y' = \varphi(y)$. Then

$$\begin{aligned} \varphi^{-1}(x'y') &= \varphi^{-1}(\varphi(x)\varphi(y)) \\ &= \varphi^{-1}(\varphi(xy)) \\ &= xy = \varphi^{-1}(x')\varphi^{-1}(y'). \end{aligned}$$

Thus φ^{-1} is a homomorphism; it is automatically bijective. ■

Definition 6.5: Automorphism group

The set of all automorphisms of G is

$$\text{Aut}(G) := \{\varphi : G \rightarrow G : \varphi \text{ is a bijective homomorphism}\}.$$

With composition of functions, $\text{Aut}(G)$ is called the *automorphism group* of G .

Proposition 6.4

$\text{Aut}(G)$ is a subgroup of the symmetric group S_G .

证明. Every automorphism is a bijection $G \rightarrow G$, so $\text{Aut}(G) \subseteq S_G$. The identity map is an automorphism. Composition of automorphisms is a homomorphism by the composition proposition and is bijective because it is a composition of bijections. The inverse of an automorphism is an automorphism by the preceding proposition. Hence $\text{Aut}(G) \leq S_G$. ■

6.5 Conjugation as an automorphism

Theorem 6.3: Conjugation is an automorphism

Let $g \in G$. The conjugation map

$$\gamma_g : G \longrightarrow G, \quad a \longmapsto gag^{-1},$$

is an automorphism. Moreover, if $H \leq G$, then

$$\gamma_g(H) = gHg^{-1} \leq G \quad \text{and} \quad H \cong gHg^{-1}.$$

证明. For $a, b \in G$,

$$\begin{aligned} \gamma_g(ab) &= gabg^{-1} \\ &= ga(g^{-1}g)bg^{-1} \\ &= (gag^{-1})(gbg^{-1}) \\ &= \gamma_g(a)\gamma_g(b). \end{aligned}$$

Thus γ_g is a homomorphism.

It is surjective: given $a' \in G$, set $a = g^{-1}a'g$. Then

$$\gamma_g(a) = g(g^{-1}a'g)g^{-1} = a'.$$

It is injective: if $\gamma_g(a) = \gamma_g(b)$, then

$$gag^{-1} = gbg^{-1}.$$

Multiplication by g^{-1} on the left and by g on the right gives $a = b$. Therefore $\gamma_g \in \text{Aut}(G)$.

If $H \leq G$, the image theorem gives $\gamma_g(H) \leq G$. The restriction

$$\gamma_g|_H : H \longrightarrow \gamma_g(H)$$

is a homomorphism, is one-to-one because γ_g is, and is onto by the definition of the image. It is an isomorphism, so $H \cong \gamma_g(H) = gHg^{-1}$. ■

The subgroup gHg^{-1} is called the *conjugate subgroup of H by g* . Conjugation can therefore produce differently situated subgroups of G , but every conjugate subgroup has exactly the same abstract group structure as H .

6.6 Homomorphisms involving the integers

6.6.1 Multiplication by an integer

Fix $m \in \mathbb{Z}$, and define

$$\varphi_m : \mathbb{Z} \longrightarrow \mathbb{Z}, \quad k \longmapsto mk.$$

Both groups are additive. For $k_1, k_2 \in \mathbb{Z}$,

$$\varphi_m(k_1 + k_2) = m(k_1 + k_2) = mk_1 + mk_2 = \varphi_m(k_1) + \varphi_m(k_2),$$

so φ_m is a homomorphism.

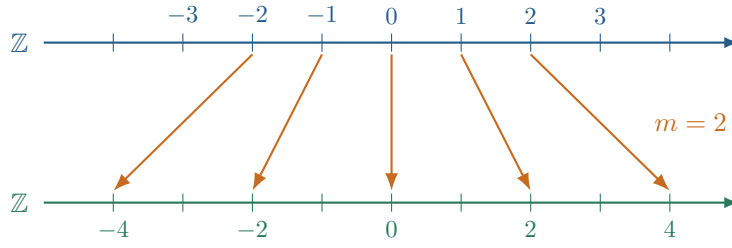


图 6.2: The homomorphism $\varphi_2(k) = 2k$. Its image contains exactly the even integers.

The complete behaviour is:

m	$\ker \varphi_m$	$\text{im } \varphi_m$	type
0	$\mathbb{Z}$	$\{0\}$	neither injective nor surjective
1	$\{0\}$	$\mathbb{Z}$	automorphism
-1	$\{0\}$	$\mathbb{Z}$	automorphism
$ m \geq 2$	$\{0\}$	$m\mathbb{Z}$	injective, not surjective

In particular, for the nontrivial nonsurjective cases,

$$\text{im } \varphi_m = m\mathbb{Z} < \mathbb{Z}.$$

6.6.2 Reduction modulo m

For $m \geq 1$, define

$$\pi_m : \mathbb{Z} \longrightarrow \mathbb{Z}/m\mathbb{Z}, \quad k \longmapsto k + m\mathbb{Z}.$$

Then

$$\begin{aligned} \pi_m(k_1 + k_2) &= (k_1 + k_2) + m\mathbb{Z} \\ &= (k_1 + m\mathbb{Z}) + (k_2 + m\mathbb{Z}) \\ &= \pi_m(k_1) + \pi_m(k_2). \end{aligned}$$

Thus π_m is a homomorphism. It is surjective because every coset has a representative $k \in \mathbb{Z}$. Its kernel is

$$\begin{aligned} \ker \pi_m &= \{k \in \mathbb{Z} : k + m\mathbb{Z} = m\mathbb{Z}\} \\ &= m\mathbb{Z}. \end{aligned}$$

For $m > 1$, it is not injective because every multiple of m maps to the identity coset.

6.7 Translations and the left regular representation

For $a \in G$, left translation is the bijection

$$\ell_a : G \rightarrow G, \quad x \longmapsto ax.$$

It is important not to confuse a bijection of the underlying set with a group homomorphism. Indeed,

$$\ell_a(xy) = axy,$$

whereas

$$\ell_a(x)\ell_a(y) = (ax)(ay) = axay.$$

If ℓ_a is a homomorphism, it must preserve the identity; hence

$$a = \ell_a(e) = e.$$

Conversely, $\ell_e = \text{id}_G$ is a homomorphism. Therefore

$$\ell_a : G \rightarrow G \text{ is a homomorphism if and only if } a = e.$$

There is nevertheless a homomorphism built from all left translations:

$$L : G \longrightarrow S_G, \quad a \longmapsto \ell_a.$$

Since

$$\ell_{ab}(x) = (ab)x = a(bx) = \ell_a(\ell_b(x)),$$

we have

$$\ell_{ab} = \ell_a \circ \ell_b \quad \text{and hence} \quad L(ab) = L(a)L(b).$$

Thus L is a homomorphism. It is injective because

$$\ell_a = \ell_b \implies a = \ell_a(e) = \ell_b(e) = b.$$

Its image is the left regular group

$$L_G = \{\ell_a : a \in G\} \leq S_G.$$

Therefore

$$L : G \longrightarrow L_G$$

is an isomorphism, and

$$G \cong L_G.$$

This is the homomorphism formulation of Cayley's theorem.

6.8 The sign homomorphism

Write a permutation $\sigma \in S_n$ in two-line notation as

$$\sigma = \begin{pmatrix} 1 & 2 & \cdots & n \\ \sigma(1) & \sigma(2) & \cdots & \sigma(n) \end{pmatrix}.$$

Definition 6.6: Sign of a permutation

The *sign* of $\sigma \in S_n$ is

$$\text{sgn}(\sigma) := \prod_{1 \leq i < j \leq n} \frac{\sigma(j) - \sigma(i)}{j - i}.$$

Equivalently,

$$\operatorname{sgn}(\sigma) = (-1)^{N(\sigma)},$$

where $N(\sigma)$ is the number of inversions, that is, pairs (i, j) with $i < j$ but $\sigma(i) > \sigma(j)$.

The numerator contains the same absolute differences as the denominator, only permuted and possibly sign-reversed. Hence the quotient is always $+1$ or -1 . A permutation is called *even* when its sign is $+1$, and *odd* when its sign is -1 .

Example 6.1

Let

$$\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}.$$

The pairs $(1, 3)$ and $(2, 3)$ are inversions, while $(1, 2)$ is not. Thus $N(\sigma) = 2$ and

$$\operatorname{sgn}(\sigma) = (-1)^2 = 1.$$

The product formula gives the same calculation:

$$\begin{aligned} \operatorname{sgn}(\sigma) &= \frac{\sigma(2) - \sigma(1)}{2 - 1} \frac{\sigma(3) - \sigma(1)}{3 - 1} \frac{\sigma(3) - \sigma(2)}{3 - 2} \\ &= \frac{3 - 2}{1} \frac{1 - 2}{2} \frac{1 - 3}{1} \\ &= 1 \cdot \left(-\frac{1}{2}\right) \cdot (-2) = 1. \end{aligned}$$

For a transposition

$$\tau = \begin{pmatrix} 1 & \cdots & i & \cdots & j & \cdots & n \\ 1 & \cdots & j & \cdots & i & \cdots & n \end{pmatrix},$$

the parity of the number of inversions is odd, so

$$\operatorname{sgn}(\tau) = -1.$$

Theorem 6.4: The sign is a homomorphism

The map

$$\operatorname{sgn} : S_n \longrightarrow \{+1, -1\}$$

is a homomorphism:

$$\operatorname{sgn}(\sigma \circ \rho) = \operatorname{sgn}(\sigma) \operatorname{sgn}(\rho).$$

证明. Insert the intermediate differences $\rho(j) - \rho(i)$:

$$\begin{aligned} \operatorname{sgn}(\sigma \circ \rho) &= \prod_{i < j} \frac{\sigma(\rho(j)) - \sigma(\rho(i))}{j - i} \\ &= \left(\prod_{i < j} \frac{\sigma(\rho(j)) - \sigma(\rho(i))}{\rho(j) - \rho(i)} \right) \left(\prod_{i < j} \frac{\rho(j) - \rho(i)}{j - i} \right). \end{aligned}$$

The second parenthesis is $\text{sgn}(\rho)$. In the first, the unordered pairs

$$\{\rho(i), \rho(j)\}, \quad i < j,$$

run exactly once through all unordered pairs of distinct elements of $\{1, \dots, n\}$. Reversing the order of a pair changes the signs of both numerator and denominator and therefore leaves their ratio unchanged. We may consequently reorder the factors with their smaller index first:

$$\prod_{i < j} \frac{\sigma(\rho(j)) - \sigma(\rho(i))}{\rho(j) - \rho(i)} = \prod_{p < q} \frac{\sigma(q) - \sigma(p)}{q - p} = \text{sgn}(\sigma).$$

Multiplying the two parentheses proves the result. ■

6.9 The alternating group

Definition 6.7: Alternating group

The *alternating group of degree n* is

$$A_n := \ker(\text{sgn}) = \{\sigma \in S_n : \text{sgn}(\sigma) = +1\}.$$

Thus A_n is the group of all even permutations.

Because a kernel is a subgroup,

$$A_n \leq S_n.$$

For $n \geq 2$, the sign map is onto: the identity has sign $+1$, while any transposition τ has sign -1 . Its two fibers are

$$A_n \quad \text{and} \quad \tau A_n.$$

Indeed, if σ is odd, then

$$\text{sgn}(\tau^{-1}\sigma) = \text{sgn}(\tau)^{-1} \text{sgn}(\sigma) = (-1)^{-1}(-1) = 1,$$

so $\tau^{-1}\sigma \in A_n$ and $\sigma \in \tau A_n$. Consequently,

$$[S_n : A_n] = 2.$$

Lagrange's theorem yields

$$|A_n| = \frac{|S_n|}{2} = \frac{n!}{2} \quad (n \geq 2).$$

6.10 Classification of cyclic groups

Theorem 6.5: Classification of cyclic groups

Let $G = \langle g \rangle$ be cyclic.

1. If G is infinite, then $G \cong \mathbb{Z}$.
2. If $|G| = m < \infty$, then $G \cong \mathbb{Z}/m\mathbb{Z} \cong \mathbb{Z}_m$.

证明. *Infinite case.* Define

$$\varphi : \mathbb{Z} \longrightarrow G, \quad k \longmapsto g^k.$$

Using additive notation on $\mathbb{Z}$,

$$\varphi(k + \ell) = g^{k+\ell} = g^k g^\ell = \varphi(k)\varphi(\ell),$$

so φ is a homomorphism. It is onto because every element of $\langle g \rangle$ is a power of g . It is one-to-one because an infinite cyclic group has no repeated powers: if $g^k = g^\ell$ with $k \neq \ell$, then $g^{k-\ell} = e$, and the powers of g would repeat with a finite positive period, making G finite. Thus φ is an isomorphism and $G \cong \mathbb{Z}$.

Finite case. Suppose $|G| = m$. Then

$$G = \{e, g, g^2, \dots, g^{m-1}\},$$

and m is the smallest positive integer for which $g^m = e$. Define

$$\psi : \mathbb{Z}/m\mathbb{Z} \longrightarrow G, \quad k + m\mathbb{Z} \longmapsto g^k.$$

We first prove that ψ is well defined. If

$$k + m\mathbb{Z} = \ell + m\mathbb{Z},$$

then $k - \ell \in m\mathbb{Z}$, so $k - \ell = qm$ for some $q \in \mathbb{Z}$. Therefore

$$g^{k-\ell} = g^{qm} = (g^m)^q = e,$$

and hence $g^k = g^\ell$.

Next,

$$\begin{aligned} \psi((k + m\mathbb{Z}) + (\ell + m\mathbb{Z})) &= \psi(k + \ell + m\mathbb{Z}) \\ &= g^{k+\ell} \\ &= g^k g^\ell \\ &= \psi(k + m\mathbb{Z})\psi(\ell + m\mathbb{Z}), \end{aligned}$$

so ψ is a homomorphism. It is onto because every element of G is a power g^k . Finally, if

$$\psi(k + m\mathbb{Z}) = \psi(\ell + m\mathbb{Z}),$$

then $g^{k-\ell} = e$. Since g has order m , this means $m \mid (k - \ell)$, and therefore

$$k + m\mathbb{Z} = \ell + m\mathbb{Z}.$$

Thus ψ is injective as well as surjective. It is an isomorphism, and

$$G \cong \mathbb{Z}/m\mathbb{Z} \cong \mathbb{Z}_m.$$

■

What to retain from this chapter

1. A homomorphism satisfies $\varphi(ab) = \varphi(a)\varphi(b)$; it preserves identities, inverses, and integer powers.
2. Images of subgroups and preimages of subgroups are subgroups. In particular, $\text{im } \varphi \leq G'$ and $\ker \varphi \leq G$.
3. A homomorphism is injective exactly when its kernel is $\{e\}$, and a composition of homomorphisms is a homomorphism.
4. Isomorphisms are bijective homomorphisms; automorphisms are isomorphisms $G \rightarrow G$, and $\text{Aut}(G) \leq S_G$.
5. Conjugation is an automorphism, and $H \cong gHg^{-1}$ for every subgroup $H \leq G$.
6. The sign map is a homomorphism $S_n \rightarrow \{\pm 1\}$, with kernel A_n , so $[S_n : A_n] = 2$ and $|A_n| = n!/2$ for $n \geq 2$.
7. Every infinite cyclic group is isomorphic to $\mathbb{Z}$, and every cyclic group of order m is isomorphic to $\mathbb{Z}_m$.